RAD51C-Related Cancer Predisposition (MONDO:0700273) — Comprehensive Disease Characteristics Report

Causal gene: ***RAD51C*** (RAD51 paralog C; HGNC:9820; NCBI Gene 5889; Ensembl ENSG00000108384; UniProt O43502; cytoband 17q22). MONDO:0700273 ("RAD51C-related cancer predisposition").

Summary

RAD51C-related cancer predisposition is an autosomal-dominant hereditary cancer syndrome caused by germline loss-of-function (LoF) variants in **RAD51C**, one of five RAD51 paralogs and an essential component of the homologous-recombination (HR) DNA double-strand-break (DSB) repair machinery and the replication-fork-protection apparatus. The disease exhibits a striking **gene-dosage duality**: a single (monoallelic/heterozygous) pathogenic variant predisposes to adult-onset tubo-ovarian and breast cancer, whereas two (biallelic) hypomorphic variants cause a **Fanconi anemia-like syndrome (Fanconi anemia complementation group O, FANCO)** with congenital malformations and bone-marrow failure.

The cancer-predisposition phenotype is dominated by a **high relative risk of high-grade serous tubo-ovarian carcinoma** (relative risk [RR] ≈ 7.5; cumulative risk to age 80 ≈ 11%) and a **moderate, triple-negative-predominant breast cancer risk** (RR ≈ 2; cumulative risk ≈ 21%). Tumors develop through a classic two-hit mechanism: a germline LoF allele plus somatic inactivation of the wild-type allele (by mutation, deletion, or—as a somatic phenocopy—promoter hypermethylation), producing an **HR-deficient ("BRCAness") tumor** with a characteristic mutational signature 3. This HR deficiency is both the oncogenic driver and the therapeutic vulnerability: RAD51C-deficient cancers are sensitive to **platinum chemotherapy and PARP inhibitors** through synthetic lethality.

Clinically, the syndrome is managed as a moderate-penetrance hereditary breast-and-ovarian cancer (HBOC) gene: **risk-reducing salpingo-oophorectomy (RRSO)** around menopause is the primary ovarian-cancer prevention, enhanced breast surveillance is offered, and genetic

counseling with cascade testing of relatives is standard. The cell of origin for the ovarian/tubal tumors is the **secretory epithelial cell of the fallopian-tube fimbria**, which acquires a *TP53* mutation forming a serous tubal intraepithelial carcinoma (STIC) before seeding the ovary and peritoneum. This report synthesizes nine confirmed findings across 65 reviewed papers into a comprehensive, section-by-section disease characterization.

Key Findings

Finding 1 — RAD51C has a dual gene-dosage phenotype

RAD51C uniquely bridges two disease categories through allele dosage. **Biallelic** (homozygous hypomorphic missense) germline mutation was first identified in a consanguineous family with multiple severe congenital abnormalities characteristic of Fanconi anemia (FA); the patient cells lost RAD51 focus formation and were hypersensitive to the crosslinking agent mitomycin C and to camptothecin, defining Fanconi anemia complementation group O (FANCO) (PMID: 20400963). As the authors state: *"We have identified a homozygous missense mutation in the RAD51C gene in a consanguineous family with multiple severe congenital abnormalities characteristic of FA."*

In contrast, **monoallelic** germline pathogenic variants are established **moderate (intermediate) penetrance** tubo-ovarian and triple-negative breast cancer predisposition alleles, as codified in the 2025 ACMG clinical practice resource (PMID: 41070818): *"RAD51C, RAD51D, and BRIP1 are moderate (intermediate) penetrance ovarian cancer predisposition genes."* This dosage duality is mechanistically coherent — complete HR failure in every cell (biallelic) is developmentally catastrophic (FA), whereas one functional allele (monoallelic) permits normal development until somatic second-hit loss triggers focal carcinogenesis.

Finding 2 — RAD51C functions as a RAD51 paralog in HR via the BCDX2 and CX3 complexes

RAD51C is a member of two RAD51-paralog subcomplexes: **BCDX2** (RAD51B–RAD51C–RAD51D–XRCC2) and **CX3** (RAD51C–XRCC3). These paralogs *"have emerged as essential tumour suppressors, forming two subcomplexes, BCDX2 and CX3"* (PMID: 39268578). BCDX2 facilitates **RAD51 loading onto single-stranded DNA (ssDNA)**, suppressing the cooperative requirement of RAD51–DNA binding and stabilizing the nucleoprotein filament—including on short ssDNA stretches and reversed replication forks.

Recent cryo-EM work reveals the mechanistic logic: paralog exchange converts an **autoinhibited RAD51–XRCC3–RAD51C octamer** into an **activated pentamer** to regulate filament formation during HR and fork preservation (PMID: 42020761). Beyond canonical DSB repair, RAD51C protects stalled replication forks; cancer-associated missense variants (**F164S, A87E, L134S, E49K**) show reduced RAD51 foci and defective fork protection, confirming that *"the role of RAD51C in DNA replication is emerging when replication forks are transiently or irreversibly stalled"* (PMID: 36562461). RAD51C also participates in R-loop resolution with the FANCM pathway, linking its FA and cancer functions (PMID: 41719405).

Finding 3 — Pathogenic variants confer high tubo-ovarian risk (RR ≈ 7.6) and moderate breast risk (RR ≈ 2)

The definitive risk estimates come from a complex segregation analysis of **6,178 families** (125 with RAD51C pathogenic variants) (PMID: 32107557): *"Pathogenic variants in both RAD51C and RAD51D were associated with TOC (RAD51C: relative risk [RR] = 7.55, 95% confidence interval [CI] = 5.60 to 10.19; P = 5 × 10-40)."* The same study established the absolute cumulative risks to age 80: *"The estimated cumulative risks of developing TOC to age 80 years were 11% (95% CI = 6% to 21%) for RAD51C."* Breast cancer RR was 1.99 (95% CI 1.39–2.85; P = 1.55×10^{-4}), with cumulative breast risk ≈ 21% (95% CI 15–29%). Notably, tubo-ovarian RR rose with age until ~60 then declined, and risks varied by family history.

The original discovery paper identified *"six monoallelic pathogenic mutations in RAD51C that confer an increased risk for breast and ovarian cancer"* (PMID: 20400964), found exclusively in breast-plus-ovarian cancer pedigrees (1.3%), absent from breast-only pedigrees and 2,912 controls.

Cancer type	Relative risk / OR	Cumulative risk to age 80	Source
Tubo-ovarian carcinoma	RR 7.55 (5.60–10.19)	11% (6–21%)	P 32107557
Breast cancer (overall)	RR 1.99 (1.39–2.85)	21% (15–29%)	P 32107557
Ovarian (meta-analysis)	OR 5.59 (4.42–7.07)	—	P 32359370
Breast (population)	OR 1.53 (1.29–2.04)	—	P 38986768

Finding 4 — Somatic RAD51C promoter hypermethylation phenocopies germline loss

Epigenetic silencing of RAD51C by promoter methylation is a somatic mechanism producing the same HR-deficient phenotype as germline mutation. Methylation is *"strongly associated with signature 3 and, in our data set, was highly enriched in basal-like breast cancers in young individuals of African descent"* (PMID: 28825726). In the phase-3 **PAOLA-1/ENGOT-ov25** trial, 25/519 (4.8%) high-grade serous ovarian cancers (HGSOC) were RAD51C-methylated; these tumors were HRD-positive and derived progression-free survival (PFS) benefit from olaparib + bevacizumab maintenance: *"A significant PFS1 benefit independently of methylation ratios was observed in patients with methylated tumors with bev-ola maintenance compared to bev alone (HR=0.49, 95 % CI 0.29-0.83, P = 0.008)"* (PMID: 40513287). The phase-2 EMBRACE trial corroborated olaparib activity in non-germline-BRCA HRD HGSOC, including RAD51C-methylated and germline RAD51C-variant cases (PMID: 42420629).

Finding 5 — Mouse models recapitulate the cancer and require Trp53 co-loss; combined Fanc mutations model FA

Epidermal/mammary *Rad51c* deletion produces mammary carcinomas *"in 63% of female mice"* **only when combined with *Trp53* inactivation**, shortening tumor latency from 11 to 6 months, and recapitulating the hormone-receptor/luminal histopathology and genomic instability of human RAD51C-mutated breast cancers (PMID: 26820992): *"deletion of Rad51c invariably requires inactivation of the Trp53 tumor suppressor (TP53 in humans) to produce mammary carcinomas in 63% of female mice."* Similarly, *Rad51c* loss accelerates *Trp53*-mutant sebaceous tumorigenesis (6 vs 11 months) but is insufficient alone (PMID: 25270124). On the FA side, combined hypomorphic *Brca2/Fancd1* + *Rad51c/Fanco* mice *"phenocopies human FA with bone marrow failure, rapid death by cancer, cellular cancer-drug hypersensitivity and severe replication instability,"* whereas single-gene inactivation is unremarkable (PMID: 36906610). The absolute requirement for *Trp53* co-loss mirrors the human *TP53*-signature STIC precursor.

Finding 6 — Saturation genome editing provides a functional map of 9,188 RAD51C variants

A landmark functional-genomics study used **saturation genome editing (SGE)** to assess *"9,188 unique variants, including >99.5% of all possible coding sequence single-nucleotide alterations"* of RAD51C (PMID: 39299233). It *"functionally classify 3,094 variants to be disruptive and use clinical truth sets to reveal an accuracy/concordance of variant classification >99.9%."* Disruptive

variants caused cell-fitness depletion, some missense alleles showed hypomorphic/intermediate kinetics, and SGE-depleted variants were significantly associated with cancer in the UK Biobank and a multicenter ovarian cohort. This map directly addresses the VUS-interpretation problem for a moderate-penetrance gene. In real-world testing cohorts, predicted LoF alterations predominate among carriers (PMID: 42322921).

Finding 7 — Pooled analyses confirm high-risk ovarian, low-frequency moderate-risk breast; RRSO is primary prevention

A meta-analysis of ~29,400 ovarian cancer cases vs ~116,000 gnomAD controls gave a RAD51C ovarian cancer **OR = 5.59 (95% CI 4.42–7.07, P < 0.0001)** (PMID: 32359370). A population-based breast cancer meta-analysis (101,397 cases / 312,944 controls) found a modest association at low pathogenic-variant frequency — *"frequency of PVs were low for RAD51C (OR 1.53, 95% CI 1.29-2.04; 1 in 913)"* — with stronger association restricted to triple-negative disease (PMID: 38986768). **Risk-reducing salpingo-oophorectomy** is the most effective ovarian prevention and is cost-effective at lifetime ovarian risk ≥ 4–5%; *"Identified carriers undergo risk-reducing salpingo-oophorectomy"* (PMID: 29361001; see also PMID: 29252925, PMID: 41070818).

Finding 8 — Tumors arise from fallopian-tube fimbrial epithelium; treated with platinum + PARP inhibitors

RAD51C-driven HGSOC shares the cell of origin of other HR-deficient tubo-ovarian cancers: *"Secretory cells in the fallopian tube fimbria epithelium (FTE) are regarded as the main cells of origin of ovarian high-grade serous carcinoma (HGSC)"* (PMID: 38956560). These secretory cells acquire *TP53* mutations, forming **serous tubal intraepithelial carcinoma (STIC)** before spreading to ovary and peritoneum; mutant p53 suppresses GSTA antioxidant defense, increasing DNA damage and genomic instability (PMID: 41275930). Therapeutically, *"The FDA has approved four PARP inhibitors (olaparib, rucaparib, niraparib, and talazoparib) for the treatment of ovarian, breast, prostate, and pancreatic cancer... approved for cancers with homologous-recombination repair deficiencies including BRCA1/2 mutations"* (PMID: 41349848). Acquired PARPi resistance arises through restored HR, fork protection, and genetic reversion (PMID: 42180878).

Finding 9 — RAD51C variants are also seen in prostate and pancreatic cancer (emerging/limited associations)

RAD51C pathogenic/likely-pathogenic variants appear in mainstream prostate cancer germline panels among DNA-damage-repair carriers: *"Fourteen patients (8%...) were found to have a deleterious germline mutation, including pathogenic or likely pathogenic variants in BRCA1/2, ATM, CHEK2, PMS2, RAD51C, HOXB13, and BRIP1"* (PMID: 38461085). In pancreatic cancer, RAD51C is classified among genes with *"a limited association or not previously associated with PC"* (PMID: 39256447). The **core established associations remain tubo-ovarian (high risk) and triple-negative breast (moderate risk)**.

Mechanistic Model / Interpretation

Ordered causal chain (germline monoallelic cancer-predisposition arm)

1. **Germline heterozygous LoF variant in RAD51C** (nonsense, frameshift, splice-site, or disruptive missense) is inherited → **reduces functional RAD51C to ~50% in every cell** (haploinsufficiency insufficient alone for tumorigenesis — one functional allele maintains HR).
2. In a fallopian-tube fimbrial secretory cell (or breast luminal progenitor), a **somatic "second hit"** — loss of heterozygosity, somatic mutation, or *promoter hypermethylation* of the wild-type allele — **results in complete loss of RAD51C function** in that clone.
3. Loss of RAD51C **disables BCDX2/CX3-mediated RAD51 nucleoprotein-filament loading onto ssDNA → abrogates homologous-recombination repair of DNA double-strand breaks** and **compromises stalled-replication-fork protection**.
4. HR deficiency **leads to error-prone compensatory repair** (NHEJ, microhomology-mediated end-joining) → **genomic instability and mutational signature 3 ("BRCAness")**.
5. *(Branch, often a prerequisite)* Co-occurring **TP53 mutation** (forming a p53-signature lesion / STIC) **removes the DNA-damage checkpoint and antioxidant defense (GSTA suppression)** → permits survival/proliferation of genomically unstable cells (mouse data show *Trp53* co-loss is essentially required).
6. Accumulated genomic instability **results in malignant transformation → serous tubal intraepithelial carcinoma (STIC)** in the fimbria → **high-grade serous tubo-ovarian**

carcinoma seeding ovary and peritoneum; in breast, a **triple-negative/basal-like carcinoma**.

7. The HR deficiency that drives the cancer **creates a therapeutic vulnerability**: **PARP inhibition** blocks single-strand-break repair, causing unrepaired DSBs that an HR-deficient cell cannot fix → **synthetic-lethal tumor-cell death**; platinum crosslinks similarly overwhelm repair capacity.
8. *(Resistance branch)* Under selective pressure, subclones **restore HR** (reversion mutations, restored fork protection) → **acquired PARP-inhibitor/platinum resistance**.

Biallelic (Fanconi anemia / FANCO) arm

1. **Two hypomorphic RAD51C alleles** inherited (often via consanguinity) → **severe constitutive HR/interstrand-crosslink-repair deficiency in all cells from conception**.
2. **Leads to defective FA–BRCA crosslink repair and R-loop resolution → developmental malformations, cellular hypersensitivity to crosslinkers (mitomycin C), chromosomal breakage, and progressive bone-marrow failure** (FANCO; demonstrated, not merely inferred, per

P 20400963

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GERMLINE LoF allele (50% RAD51C)
         │
         ▼  somatic 2nd hit (LOH / mutation / promoter methylation)
Complete RAD51C loss in a fimbrial secretory cell
         │
         ▼  BCDX2/CX3 cannot load RAD51 → HR fails + fork protection lost
HR DEFICIENCY ("BRCAness", signature 3)
         │  + TP53 mutation (STIC precursor)
         ▼
Genomic instability → STIC → HIGH-GRADE SEROUS TUBO-OVARIAN CARCINOMA
         │                          └─(breast arm)→ triple-negative/basal carcinoma
         ▼  THERAPEUTIC EXPLOIT
PARP inhibitor / platinum → synthetic lethality → tumor death
         │
         ▼  (resistance) HR restoration / reversion → relapse
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Ontology term suggestions

- **Gene/protein:** HGNC:9820 (RAD51C); GO:0000724 (DSB repair via HR); GO:0006310 (DNA recombination); GO:0036297 (interstrand cross-link repair); GO:0031297 (replication fork

processing); GO:0005654 (nucleoplasm, CC); CHEBI:63631 (olaparib); CHEBI:27899 (cisplatin).

- **Phenotypes (HPO):** HP:0100615 (ovarian neoplasm); HP:0003002 (breast carcinoma); HP:0012125 (prostate cancer); HP:0002664 (neoplasm); HP:0005528 (bone marrow hypocellularity, FA arm); HP:0001510 (growth delay, FA arm); HP:0040012 (chromosome breakage).
- **Cell types (CL):** CL:0002536 (fallopian tube secretory epithelial cell); CL:0000066 (epithelial cell); CL:0002326 (luminal epithelial cell of mammary gland).
- **Anatomy (UBERON):** UBERON:0003889 (fallopian tube); UBERON:0000996 (oviduct infundibulum/fimbria); UBERON:0000992 (ovary); UBERON:0002367 (peritoneum); UBERON:0000310 (breast).
- **Disease (MONDO):** MONDO:0700273 (RAD51C-related cancer predisposition); related MONDO:0011705 (Fanconi anemia complementation group O).
- **Treatment (NCIT):** NCIT:C71721 (Olaparib); NCIT:C78844 (Rucaparib); NCIT:C97660 (Niraparib); NCIT:C95733 (Talazoparib); NCIT:C51686 (salpingo-oophorectomy); NCIT:C1282 (Carboplatin).

Section-by-Section Disease Characterization

1. Disease Information

RAD51C-related cancer predisposition (MONDO:0700273) is an **autosomal-dominant HBOC-spectrum syndrome** caused by heterozygous germline LoF variants in **RAD51C**. Key identifiers: **Gene** RAD51C (HGNC:9820, NCBI Gene 5889, chr17q22); **MONDO:0700273**; related OMIM — RAD51C gene *613390*, Fanconi anemia group O *613390/614082*, breast-ovarian cancer susceptibility (BROVCA3) *613399*. Synonyms/related: "RAD51C-associated hereditary breast and ovarian cancer," "FANCO (biallelic)," "BROVCA3." Information is derived from **aggregated disease-level resources** (segregation analyses, case-control meta-analyses, functional assays) rather than individual EHR.

2. Etiology

Primary cause: germline heterozygous LoF variant in RAD51C (genetic). **Genetic risk factors:** pathogenic RAD51C variants (RR ovarian 7.55, breast 1.99; P 32107557); family history modifies risk. **Second-hit** somatic events (LOH, promoter hypermethylation) are required for tumorigenesis. **Environmental/gene-environment:** limited specific data; general HR-targeting carcinogens such as hexavalent chromium transcriptionally repress RAD51 paralogs in vitro (PMID: 42169974) — mechanistically plausible as an exogenous phenocopy but not an established disease modifier. **Protective factors:** RRSO and (by analogy to BRCA) oral contraceptives for ovarian risk; no validated protective genetic modifier alleles identified. Consanguinity is a risk factor for the biallelic FA arm.

3. Phenotypes

Phenotype	Type	Onset	Frequency / risk	HPO
High-grade serous tubo-ovarian carcinoma	Clinical sign / neoplasm	Adult (risk rises to ~60y)	Cumulative ~11% by 80; RR 7.55	HP:0100615
Triple-negative / basal breast carcinoma	Neoplasm	Adult	Cumulative ~21%; RR ~2	HP:0003002
Prostate cancer (emerging)	Neoplasm	Adult/older	Among DDR-panel carriers	HP:0012125
Bone-marrow failure (biallelic FANCO)	Lab/clinical	Congenital/ childhood	Biallelic cases	HP:0005528
Congenital malformations (biallelic)	Physical	Congenital	Biallelic cases	HP:0001510

Severity is **variable** (moderate penetrance monoallelic; severe biallelic). Quality-of-life impact derives chiefly from cancer treatment and from surgical menopause after RRSO.

4. Genetic/Molecular Information

Causal gene: RAD51C (HGNC:9820). **Variant classes:** predominantly LoF — nonsense, frameshift, splice-site, large deletions; plus disruptive missense (e.g., F164S, A87E, L134S, E49K with defective fork protection, (P 36562461).

Classification: ACMG/AMP pathogenic/likely-pathogenic/VUS; SGE map of 9,188 variants classifies 3,094 disruptive with >99.9% concordance (P 39299233).

Allele frequency: pathogenic-variant carrier frequency ≈ **1 in 913** in population breast cohorts (P 38986768) — rare. **Somatic vs germline:** both; **somatic promoter hypermethylation** is a recognized epigenetic LoF mechanism (P 28825726).

Functional consequence: loss of function (tumor-suppressor). **Epigenetic:** RAD51C promoter CpG hypermethylation silences expression and produces HRD/signature 3.

5. Environmental Information

No established infectious or specific environmental cause. In vitro, **hexavalent chromium [Cr(VI)]** transcriptionally represses all RAD51 paralogs and inhibits RAD51 filament formation (P 42169974) — a mechanistic parallel, not an epidemiologically proven disease risk factor. Lifestyle factors are not specifically established for RAD51C beyond general cancer-risk considerations.

6. Mechanism / Pathophysiology

Covered in detail in the **Mechanistic Model** above. Core pathway: **HR DSB repair (GO:0000724)** via the **BCDX2** and **CX3** RAD51-paralog complexes, plus **replication-fork protection**; loss → genomic instability → malignant transformation gated by TP53 loss. Cell of origin: fallopian-tube fimbrial secretory cells (CL:0002536). Subcellular compartment: nucleus/replication fork.

7. Anatomical Structures Affected

- **Primary organs:** fallopian tube (fimbria; UBERON:0003889/0000996), ovary (UBERON:0000992), breast (UBERON:0000310).
- **Secondary:** peritoneum (UBERON:0002367) via transcoelomic spread; prostate (emerging).
- **Body systems:** female reproductive; hematopoietic in biallelic FA.
- **Cell/tissue:** epithelial (secretory tubal cell, mammary luminal progenitor).
- **Lateralization:** breast often unilateral at presentation but with bilateral risk; tubo-ovarian can be bilateral.

8. Temporal Development

Adult-onset cancer predisposition. Tubo-ovarian **RR increases with age until ~60 then decreases** (P 32107557). Progression follows HGSOC natural history: STIC → invasive HGSC → advanced (stage III/IV) peritoneal disease, typically presenting late. Biallelic FANCO is congenital/childhood-onset and progressive (bone-marrow failure). Critical intervention window: **RRSO near menopause** (before cancer onset).

9. Inheritance and Population

- **Inheritance:** autosomal dominant (monoallelic cancer predisposition); autosomal recessive (biallelic FANCO).
- **Penetrance:** incomplete, age-dependent, moderate.
- **Carrier frequency:** ~1 in 913 (breast cohorts).
- **Populations:** reported across global cohorts; RAD51C methylation enriched in basal-like breast cancers of young women of African descent (P 28825726); founder-type recurrences occasionally reported.
- **Sex:** cancer burden overwhelmingly female (breast/ovarian); prostate in males.

10. Diagnostics

- **Genetic testing:** multigene HBOC panels including RAD51C (NGS); single-gene/cascade testing for relatives. SGE functional map aids VUS interpretation (P 39299233).
- **Tumor HRD testing:** mutational signature 3, genomic-instability scores, RAD51C methylation assays (P 40513287); RAD51C immunohistochemistry proposed as part of composite HRD biomarker (P 39402170).
- **Pathology:** high-grade serous histology, abnormal p53 (STIC); triple-negative/basal breast.
- **Differential:** BRCA1/2, RAD51D, BRIP1, PALB2, ATM, CHEK2, BARD1-related HBOC.

11. Outcome / Prognosis

Prognosis tracks HGSOC/triple-negative breast natural history. **HRD status is favorable for therapy response:** PARPi maintenance yields large PFS gains (olaparib+bevacizumab HR 0.49 in methylated tumors, P 40513287; broad PARPi PFS benefit in HRD, Cochrane P 43012369). Acquired resistance via HR restoration limits durability (P 42180878). Prognostic biomarker: HRD/signature 3 predicts platinum/PARPi sensitivity.

12. Treatment

- **Pharmacotherapy / targeted: PARP inhibitors** — olaparib (NCIT:C71721), rucaparib (NCIT:C78844), niraparib (NCIT:C97660), talazoparib (NCIT:C95733) — FDA-approved for HRD cancers (P 41349848); **platinum** chemotherapy (carboplatin/cisplatin).

- **Surgical/preventive:** risk-reducing salpingo-oophorectomy (NCIT:C51686); risk-reducing mastectomy considered case-by-case (generally not mandated for an isolated moderate-risk breast variant).
- **Combination/experimental:** PARPi + bevacizumab; PARPi + immune checkpoint inhibitors under study
(P 41541937).
- **Pharmacogenomics/personalized:** genotype/HRD-guided PARPi selection.

13. Prevention

- **Primary (ovarian):** RRSO near menopause — most effective, cost-effective at lifetime risk ≥4–5%
(P 29252925). P 29361001
- **Secondary:** enhanced breast surveillance (intensified imaging, e.g., MRI); no effective ovarian screening exists.
- **Genetic counseling / cascade testing** of at-risk relatives; prenatal/PGT options for biallelic-risk families.
- **Risk stratification:** multigene panels; population-based testing cost-effective
(P 29361001).

14. Other Species / Natural Disease

RAD51C is evolutionarily conserved (mouse *Rad51c*, NCBI Gene 114714). No prominent naturally occurring companion-animal disease documented in this investigation; the gene's essentiality means homozygous knockout is embryonic-lethal in mouse.

15. Model Organisms

- **Mouse (conditional knockouts):** *Rad51c* deletion + *Trp53* loss → mammary carcinoma (63% of females; latency 11→6 months) recapitulating human luminal/HR-deficient pathology
(P 26820992);

Rad51c/Trp53 sebaceous-tumor model
( ).
P 25270124

- **FA model:** hypomorphic *Brca2/Fancd1* + *Rad51c/Fanco* double mutants phenocopy human Fanconi anemia (bone-marrow failure, cancer, replication instability)
( ).
P 36906610
- **In vitro / functional genomics:** saturation genome editing in human cells
( );
P 39299233
biochemical/cryo-EM reconstitution of BCDX2/CX3
(
P 39268578
).
P 42020761
- **Limitation:** single-gene *Rad51c* loss alone is insufficient for tumorigenesis — the obligate *Trp53* requirement is both a strength (models human STIC genetics) and a caveat for isolating RAD51C-specific effects.

Evidence Base

PMID	Title (abbrev.)	Role
20400963	*RAD51C mutation in Fanconi anemia-like disorder*	Establishes biallelic → FANCO
20400964	*Germline mutations establish RAD51C as cancer gene*	Original cancer-gene discovery
32107557	*Ovarian/breast risks in RAD51C/RAD51D*	Definitive RR/cumulative-risk estimates
32359370	*Pooled ~30,000-case ovarian analysis*	Meta-analytic ovarian OR 5.59
38986768	*Population breast meta-analysis*	Breast OR 1.53; PV freq 1/913
39268578	*BCDX2 RAD51 nucleation mechanism*	Molecular HR function
42020761	*RAD51 paralog exchange cryo-EM*	Structural regulation of filament
36562461	*RAD51C missense variants & fork protection*	Replication-fork role; specific variants
28825726	*Signature 3 / RAD51C methylation*	Epigenetic LoF → HRD
40513287	*PAOLA-1 methylation ancillary study*	Methylation predicts PARPi benefit
42420629	*EMBRACE phase 2 olaparib*	PARPi in non-gBRCA HRD
26820992	*Rad51c/Trp53 mouse model*	Tumor model; Trp53 requirement
36906610	*Brca2/Rad51c double-mutant mice*	FA mouse model
25270124	*Rad51c sebaceous tumor model*	Synergy with Trp53
39299233	*Saturation genome editing of RAD51C*	Functional variant map
41070818	*ACMG clinical practice resource*	Management/penetrance guidance
29361001	*Cost-effectiveness population testing*	RRSO as prevention
29252925	*Surgical prevention thresholds*	RRSO cost-effectiveness ≥4–5%
38956560	*Fallopian tube cell of origin*	FTE secretory cell origin
41275930	*Mutant p53 in tubal carcinogenesis*	TP53/STIC branch
41349848	*FDA-approved PARP inhibitors*	Four approved PARPi

PMID	Title (abbrev.)	Role
42180878	*PARPi resistance review*	HR-restoration resistance
38461085	*Prostate mainstream testing*	RAD51C in prostate panels
39256447	*Pancreatic germline variants*	Limited pancreatic association

Note on a contrary study: a large exome-sequencing case-control study (PMID: 30128536) confirmed the RAD51C ovarian association (FDR-corrected P = .004) but *did not support* a RAD51C breast cancer association, underscoring that the breast signal is weaker and largely concentrated in triple-negative disease.

Limitations and Knowledge Gaps

- **No primary dataset was analyzed** in this investigation; findings are a literature synthesis of aggregated disease-level evidence. Statistical estimates are quoted from published cohorts, not independently recomputed.
- **Penetrance uncertainty:** cumulative-risk confidence intervals are wide (ovarian 6–21%), and risk is family-history-dependent; the breast cancer association is contested (P 30128536) and concentrated in triple-negative disease.
- **VUS burden:** despite the SGE map, missense interpretation remains challenging, especially for hypomorphic/intermediate alleles.
- **Modifier genes and gene–environment interactions** for RAD51C are largely uncharacterized; the Cr(VI) link is in vitro only.
- **Prostate and pancreatic risk** are emerging/limited and not quantified with the rigor of ovarian risk.
- **Model limitation:** obligate *Trp53* co-loss in mouse models complicates isolation of RAD51C-specific tumor biology.
- **Resistance biology:** mechanisms of acquired PARPi/platinum resistance specific to RAD51C (vs BRCA) are incompletely defined.

Proposed Follow-up Experiments / Actions

1. **Refine age-specific penetrance** for RAD51C by sex and cancer subtype using large prospective biobanks (UK Biobank, All of Us), enabling individualized RRSO timing.
2. **Integrate the SGE functional map into ClinVar/VarSome pipelines** to reclassify RAD51C VUS and prospectively validate against carrier outcomes.
3. **Quantify prostate and pancreatic RAD51C risk** via dedicated case-control meta-analyses to determine whether surveillance guidelines should expand.
4. **Prospectively evaluate RAD51C methylation and IHC as predictive HRD biomarkers** for PARPi selection in HGSOC, building on PAOLA-1 ancillary data.
5. **Dissect RAD51C-specific PARPi resistance** (reversion vs fork-protection restoration) in patient-derived organoids to design rational combination therapy (e.g., PARPi + ATR/CHK1 inhibitors).
6. **Model the biallelic FANCO arm** and R-loop/FANCM axis to clarify shared versus divergent mechanisms with the cancer-predisposition arm.
7. **Evaluate chemoprevention and non-surgical ovarian-risk reduction** (e.g., salpingectomy with delayed oophorectomy) in RAD51C carriers to mitigate surgical-menopause morbidity.